

The Loop Danger

Q26

REACTIVITY

INTERVIEW QUESTION

Question. How does reactivity work traditionally, and what is the danger when an effect modifies a state it depends on?

ANSWER. In Svelte 3 and 4, reactivity is achieved by looking for variable assignments. At compile time, when the compiler sees an assignment like `count += 1`, it invisibly wraps that assignment in an invalidate function that immediately tells the precise associated DOM node to update directly.

App.svelte

```
<script>
  // Svelte 4 Reactivity
  let count = 0;

  function add() {
    count += 1; // Triggers UI update
  }
</script>

<button on:click={add}>
  Clicks: {count}
</button>
```

`count = 1`



COMPILER



`$$invalidate('count', ...)`

DOM UPDATES DIRECTLY

- ① **Summary.** Reactivity in older Svelte versions relies on explicit variable assignments. The compiler statically analyzes the code and injects direct DOM updates wherever it detects these assignments, completely bypassing the need for a Virtual DOM.